

Joel Iziren

Software Engineer (Frontend & Systems)

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PROFESSIONAL SUMMARY

Frontend-focused Software Engineer with experience building responsive web applications and low-level systems for blockchain data processing. Skilled in React, Next.js, and Tailwind CSS, with hands-on experience in Bitcoin transaction parsing, chain analysis, and data visualization. Strong understanding of transaction mechanics, data structures, and building end-to-end systems without reliance on external APIs.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript
Frameworks	React.js, Next.js, Node.js, Tailwind CSS, Formik
Tools & Others	Git, GitHub, Cypress, Appium, JMeter, Browser DevTools, Wireless Android Debugging
Concepts	REST APIs, Blockchain Fundamentals, Data Structures, Responsive Design

PROFESSIONAL EXPERIENCE

ALX Software Engineering Program

Sept 2023 - Nov 2024

Software Engineering Intern

Frontend Specialization

- Completed an intensive program covering frontend development, backend fundamentals, and software testing.
- Built reusable UI components and dynamic pages (Home, Report, View Missing Persons, Login, Signup) for a production-style application.
- Implemented form handling and validation with Formik, reducing input errors and improving user interaction reliability.
- Applied Tailwind CSS and responsive design principles to ensure consistency across devices.
- Maintained structured Git workflows with clear commit history, improving collaboration and version control.
- Gained hands-on experience with automated testing tools including Cypress, Appium, and JMeter.

PROJECTS

Summer of Bitcoin Challenge

2026

Bitcoin Chain Analysis Suite

Backend, Systems, Data Analysis

- Built a Bitcoin transaction parser to decode raw blockchain data from `blk.dat`, `rev.dat`, and `xor.dat` files without relying on external APIs or full nodes.
- Implemented a PSBT (Partially Signed Bitcoin Transaction) builder to support secure transaction construction and serialization workflows.
- Developed a chain analysis engine using heuristics such as Common Input Ownership, change detection, CoinJoin detection, and address reuse.
- Designed transaction classification logic to categorize activity into simple payments, CoinJoins, consolidations, batch payments, and self-transfers.
- Generated structured JSON reports and Markdown summaries for multi-block datasets to support reproducibility and automated evaluation.
- Built an interactive web dashboard to visualize transaction metrics including fee rates, script types, and heuristic signals.
- Integrated block file uploads via UI while preserving analysis pipelines and dataset consistency.
- Validated fee calculations, virtual size (vsize), witness parsing, and undo-data alignment at a low level.

Amber Alert PRO

2024

Team Project

Frontend & Fullstack Development

- Built a fullstack web application for reporting missing persons and enabling real-time alert distribution.
- Developed reusable UI components and dynamic pages using React, Next.js, Tailwind CSS, and Formik.
- Implemented backend services with Node.js, Express.js, MongoDB, and Firebase Authentication.
- Integrated Google Maps API for location-based reporting and visualization.

EDUCATION

University of Abuja
BSc. Computer Science

2022

Data Structures, Algorithms, Computer Networks